Medical Image Classification Module

MedAssistAI

MILESTONE-2

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1. Project Description

Symptom-Based Disease Prediction Module

The objective of this module is to develop an AI-based disease prediction system capable of identifying probable diseases using patient symptom information. The module is a core component of the MedAssistAI project and complements the Medical Image Classification Module by providing symptom-based diagnosis before image analysis.

A reusable machine learning pipeline was developed to preprocess symptom data, train prediction models, evaluate their performance, and save trained models for future deployment. The pipeline includes data cleaning, label encoding, train-test splitting, model training, prediction, and evaluation using a common workflow.

During this milestone, **LightGBM** was selected as the baseline machine learning model because of its effectiveness on structured tabular datasets. The trained model was evaluated using standard classification metrics. Although the obtained accuracy was low due to the complexity of the dataset, the preprocessing pipeline and evaluation framework were successfully completed. The developed workflow can now be reused to evaluate additional machine learning models such as Random Forest, XGBoost, and CatBoost.

2. Objective

The The objectives of this work were:

- Develop a reusable symptom-based disease prediction pipeline.
- Perform preprocessing of the symptom dataset.
- Prepare the dataset for multiclass machine learning.
- Train a baseline LightGBM classification model.
- Evaluate model performance using standard classification metrics.
- Analyze dataset characteristics affecting prediction performance.
- Save the trained model for future deployment.
- Establish a reusable framework for evaluating additional machine learning models.
- Support integration of the prediction module with the MedAssistAI system.

3. Dataset Used

The Symptom-Based Disease Prediction module utilizes the **Disease and Symptoms Preprocessed Dataset**, which contains patient symptom information represented in a structured binary format. Each row in the dataset corresponds to a patient record, where symptoms are encoded as binary features indicating the presence (1) or absence (0) of a particular symptom. The target variable represents the diagnosed disease associated with the reported symptoms.

The dataset consists of **189,601 patient records**, **377 symptom features**, and **727 disease classes**, making it a large-scale multiclass classification problem. The symptom features include common clinical indicators such as fever, cough, headache, chest pain, fatigue, vomiting, dizziness, and many others that are commonly observed during preliminary medical examinations.

Before model training, the dataset underwent preprocessing to remove invalid disease classes, standardize feature names, encode disease labels, and prepare the data for machine learning. This preprocessing ensured that the dataset was suitable for training and evaluating the prediction model.

Dataset Summary

Parameter	**Value**
Dataset Name	Disease and Symptoms Preprocessed Dataset
Total Records	189,601
Symptom Features	377
Disease Classes	727
Feature Type	Binary (0 = Absent, 1 = Present)
Target Variable	Disease Name
Prediction Task	Multiclass Disease Classification

4. Environment Setup

The Symptom-Based Disease Prediction module was implemented using **Python** in a **Jupyter Notebook** environment. Python was chosen due to its extensive support for data preprocessing, machine learning, and model evaluation. Various open-source libraries were utilized to simplify data analysis, visualization, model training, and performance evaluation.

The baseline classification model was developed using the **LightGBM** library. Additional libraries such as **Pandas** and **NumPy** were used for data manipulation, while **Scikit-learn** was used for preprocessing, train-test splitting, label encoding, and evaluation metrics. Model serialization was performed using the **Joblib** library to enable future deployment within the MedAssistAI system.

5. Dataset Analysis

Before training the machine learning model, the dataset was analyzed to understand its characteristics and identify potential challenges that could affect model performance. The analysis revealed that the dataset contains **189,601 patient records**, **377 symptom features**, and **727 disease classes**, making it a highly complex multiclass classification problem.

Further analysis showed that the dataset suffers from **class imbalance**, where some diseases have thousands of patient records while many others have only a few samples. Such an uneven distribution makes it difficult for the model to learn all disease classes effectively and can lead to biased predictions toward frequently occurring diseases.

The dataset also contained duplicate symptom combinations and conflicting records, where identical symptom patterns were associated with different diseases. These inconsistencies increase the complexity of the prediction task and negatively impact the overall classification performance. Understanding these characteristics helped in designing an appropriate preprocessing pipeline and selecting suitable machine learning algorithms for further experimentation.

6. Data Preprocessing

Data preprocessing is an essential step in developing a reliable machine learning model. The raw symptom dataset was cleaned and transformed to ensure that it was suitable for training and evaluation.

Initially, the feature names were standardized by removing special characters and formatting inconsistencies. This ensured compatibility with machine learning libraries and simplified further data processing.

The disease labels were then analyzed, and classes with fewer than two samples were removed. This was necessary to perform stratified train-test splitting, ensuring that each disease class was represented in both the training and testing datasets.

Since machine learning models require numerical target values, the disease names were converted into numerical labels using **Label Encoding**. Finally, the dataset was divided into **80% training data** and **20% testing data** using stratified sampling to preserve the original distribution of disease classes.

The preprocessing pipeline produced a clean and structured dataset that could be directly used for model training and performance evaluation.

Preprocessing Steps

1. Loaded the preprocessed symptom dataset.
2. Standardized feature (column) names.
3. Removed disease classes with fewer than two samples.
4. Encoded disease labels into numerical values.
5. Split the dataset into training and testing sets using stratified sampling.
6. Prepared the processed data for model training.

The preprocessing stage ensured data consistency, improved compatibility with the machine learning algorithm, and established a reusable pipeline for evaluating multiple classification models in future experiments.

7. Proposed System

The proposed system is designed to predict diseases based on the symptoms provided by the user. The system follows a machine learning pipeline that begins with loading the preprocessed symptom dataset, followed by data preprocessing, model training, prediction, and performance evaluation. A LightGBM classifier was used as the baseline model to learn the relationship between symptom patterns and disease classes.

After training, the model predicts the most probable disease for a given set of symptoms. The prediction results are evaluated using standard performance metrics such as Accuracy, Precision, Recall, and F1-Score. Finally, the trained model and label encoder are saved to enable future deployment and integration into the MedAssistAI application.

System Workflow:

Dataset
↓
Preprocessing
↓
Feature Engineering
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LightGBM
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Prediction
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Evaluation
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Model Saving

Modules of the Proposed System

- **Dataset Loading:** Reads the preprocessed symptom dataset.
- **Data Preprocessing:** Cleans and prepares the dataset for training.
- **Model Training:** Trains the LightGBM classifier using the training dataset.
- **Prediction:** Predicts the disease based on the given symptoms.
- **Evaluation:** Measures the model's performance using classification metrics.
- **Model Saving:** Stores the trained model and label encoder for future use.

The modular design of the system allows additional machine learning algorithms, such as Random Forest, XGBoost, and CatBoost, to be integrated into the same pipeline for performance comparison in future work.

8. Training Pipeline

The processed dataset was used to train a **LightGBM** classifier, which was selected as the baseline model due to its efficiency in handling large structured datasets. The model was trained using the training dataset and learned the relationship between symptom patterns and disease classes.

After training, the model was tested using the unseen test dataset. The trained model and label encoder were then saved using the **Joblib** library for future prediction and deployment in the MedAssistAI system.

9. Model Evaluation

The performance of the LightGBM model was evaluated using standard classification metrics, including Accuracy, Precision, Recall, and F1-Score.

Evaluation Results

Metric	Value
Accuracy	1.48%
Precision	2.72%
Recall	1.48%
F1-Score	0.75%

The model achieved low prediction performance because the dataset contains **727 disease classes** with significant class imbalance and many diseases having only a few samples. Despite the low accuracy, the experiment established a baseline for comparing more advanced machine learning models in future work.

10. Challenges & Solutions

Challenge	Solution
Large number of disease classes	Used LightGBM as a baseline model for initial experimentation.
Class imbalance	Removed disease classes with fewer than two samples and used stratified train-test splitting.
Duplicate and conflicting records	Analyzed dataset quality before model training.
Low prediction accuracy	Planned to evaluate additional machine learning models in future milestones.

12. Conclusion

In this milestone, a complete symptom-based disease prediction pipeline was successfully developed. The dataset was preprocessed, analyzed, and prepared for machine learning, followed by training and evaluation of a LightGBM classifier. Although the model achieved low accuracy due to the complexity and imbalance of the dataset, the preprocessing workflow and experimental framework were successfully established. This milestone provides a strong foundation for evaluating more advanced models and integrating symptom-based disease prediction into the MedAssistAI system.